

Incidence of Radiation Retinopathy after High-dosage Single-fraction Gamma Knife Radiosurgery for Choroidal Melanoma

Anton Haas, MD,¹ Oliver Pinter, MD,¹ Georg Papaefthymiou, MD,² Martin Weger, MD,¹ Andrea Berghold, PhD,³ Oskar Schröttner, MD,² Klaus Müllner, MD,¹ Gerhard Pendl, MD,² Gerald Langmann, MD¹

Objective: To investigate the incidence and clinical findings of radiation retinopathy after single-fraction high-dose gamma knife radiosurgery for choroidal melanoma.

Design: Retrospective noncomparative interventional case series.

Participants: Thirty-two patients with choroidal melanoma.

Methods: Review of charts, color fundus photographs, and fluorescein angiograms of 32 choroidal melanoma patients after radiosurgery. All patients were treated with the Leksell gamma knife in one fraction with a marginal dose between 40 and 80 Gy (median, 50 Gy) and were followed for at least 24 months (or until enucleation because of complications secondary to radiation).

Main Outcome Measures: Any clinical feature of radiation retinopathy and neovascular glaucoma.

Results: During a mean follow-up of 38 months (range, 6–81 months) we found radiation retinopathy in 84% of our patients. The most common findings in these patients were intraretinal hemorrhages with an incidence of 70%, macular edema and capillary nonperfusion in 63%, and hard exudates in 52% of the patients. Less common were microaneurysms in 30% and retinal neovascularization in 22%. The time of onset of the various radiation-associated retinal findings ranged between 1 and 22 months. Forty-seven percent of all patients developed neovascular glaucoma. In our study there was no correlation between radiation dosage applied and clinical findings.

Conclusions: Single-fraction high-dose Leksell gamma knife radiosurgery of choroidal melanomas with a median marginal dose of 50 Gy is highly associated with early radiation retinopathy and with neovascular glaucoma. *Ophthalmology* 2002;109:909–913 © 2002 by the American Academy of Ophthalmology.

Plaque radiotherapy and proton beam irradiation have largely replaced enucleation as treatment for choroidal melanoma.^{1,2} Newer treatment modalities such as stereotactic radiosurgery with the gamma knife, transpupillary thermotherapy, or local resection have also proven to be effective in the control of tumor growth.^{3–6} All these treatments, however, have complications. Radiation retinopathy is common after ophthalmologic radiotherapy. An accurate estimate of the incidence is difficult, because such variables as the radiation delivery system, radiation dose, location of the tumor, and tumor size influence the development of radiation retinopathy. Radiation retinopathy has been reported in 23% to 43% of patients after iodine-125 plaque radiother-

apy and in 34% of patients after proton beam irradiation.^{7–9} In a study of patients with juxtapapillary melanoma, radiation retinopathy was found in 87%.¹⁰

Radiation retinopathy is a slowly progressive occlusive microangiopathy involving the retinal vasculature. The damage to the vasculature appears as intraretinal hemorrhages, microaneurysms, cotton-wool spots, vascular shedding, capillary nonperfusion, and lipid exudates. The more advanced or late stages of radiation retinopathy feature such vision- or eye-threatening changes as neovascular glaucoma, retinal neovascularization, vitreous hemorrhages, and traction retinal detachment. Radiation doses in the 45 to 55 Gy range delivered to half or more of the retina are likely to produce radiation retinopathy.¹¹ The threshold for radiation retinopathy also depends on the fractionation scheme, the radiation delivery system, the size of the tumor treated, and the presence of such systemic risk factors as diabetes mellitus.¹²

When the gamma knife was first used to treat choroidal melanomas in the early 1990s, the radiation doses were chosen to produce tumor regression rather than to avoid side effects, and marginal doses ranged from 45 to 80 Gy. It has since been recognized that such side effects as radiation retinopathy occur as a function of radiation dose and tumor

Originally received: December 29, 2000.

Accepted: October 9, 2001.

Manuscript no. 200934.

¹ Department of Ophthalmology, Karl-Franzens University, Graz, Austria.

² Department of Neurosurgery, Karl-Franzens University, Graz, Austria.

³ Institute of Medical Informatics, Statistics and Documentation, Karl-Franzens University, Graz, Austria.

Presented in part at the 22nd Meeting of the Club Jules Gonin, Toarmina, Italy, September 2000.

Reprint requests to Anton Haas, MD, Department of Ophthalmology, Karl-Franzens University, Auenbruggerplatz 4, 8036 Graz, Austria.

size and must be assessed when the gamma knife is used as a treatment option for choroidal melanomas. The aim of this study was to investigate the onset, incidence, and clinical picture of radiation retinopathy after single-fraction gamma knife radiosurgery for choroidal melanoma.

Patients and Methods

A retrospective study reviewed the medical records, color fundus photographs, and fluorescein angiograms of 32 patients who underwent surgery for choroidal melanomas with the Leksell gamma knife (Elekta Instrument AB, Stockholm, Sweden) from June 1992 through October 1997. They were followed for at least 24 months or until enucleation because of complications from radiation retinopathy. Baseline patient data included gender, age, and any history of diabetes mellitus, systemic hypertension, or current chemotherapy. All patients underwent full ophthalmologic assessment before treatment, including best-corrected visual acuity, intraocular pressure, direct and indirect ophthalmoscopy, color fundus photography, fluorescein and indocyanine green angiography, and A- and B-scan ultrasonography. Indocyanine green angiography was performed routinely to document tumor vessels and to follow tumor regression but not to document radiation retinopathy. Follow-up examinations were at 2 weeks, and 1, 3, 6, 9, and 12 months and then at 3- to 6-month intervals thereafter. The choroidal melanoma data consisted of tumor thickness shown by A- and B-scan ultrasonography, maximum diameter of the tumor base and location of the tumor.

Nonproliferative radiation retinopathy was diagnosed if any of the following features was documented: microaneurysms, intraretinal hemorrhages, macular edema, lipid exudates, capillary nonperfusion, vascular sheathing, and cotton-wool spots. The diagnosis of proliferative radiation retinopathy depended on the presence of retinal or optic disc neovascularization. Such additional complications of radiation retinopathy as neovascular glaucoma and vitreous hemorrhage were documented. The development of these clinical features and the time to their onset were documented by direct and indirect ophthalmoscopy, color fundus photography, and fluorescein angiography.

Treatment

All patients were treated with the Leksell gamma knife, whereby gamma radiation is emitted from 201 cobalt sources (cobalt-60) localized in a hemisphere around the patient's head. The beam converges from all 201 sources to a focal point. Before radiosurgery the Leksell stereotactic frame (Elekta Instrument AB) was fixed with four pins at the tabula externa with the patient under local anesthesia. In contrast to common neurosurgical procedures, for treatment of retinal structures the patient's head is positioned in an oblique manner with a minimum distance between the frame and the back of the head. The pins are fixed in a supraorbital position to lower the z-value, and positioning of the eccentrically located eye in the focus of the gamma unit is possible. To immobilize the eye, all four rectus muscles were fixated transconjunctivally, and the sutures were firmly attached to the stereotactic frame under a retrobulbar anesthetic block of 2.5 ml mepivacaine and 2.5 ml bupivacaine. This procedure guaranteed immobilization of the eye during the preparatory and treatment phases.

Ocular structures and the stereotactic frame were imaged with computed tomography or magnetic resonance imaging. The Kula software package (Elekta Instrument AB) was used for dose planning in our first eight patients; since 1994, we have used the Leksell GammaPlan. Critical structures such as the optic nerve, the

cornea and the lens were shown on the images of the Leksell GammaPlan. Treatment planning volume was defined as the gross tumor volume plus a safety margin of 1 to 1.5 mm around the tumor. The patients were treated with a marginal dose between 40 and 80 Gy and a maximal dose between 70 and 125 Gy in a single fraction. Collimator channels are plugged to minimize radiation damage to such radiosensitive structures as the lens or optic nerve and to prevent any radiation from penetrating the other eye. Radiosurgery was performed in either the prone or supine position; 8-mm collimators or, occasionally, 12-mm collimators were used. The exposure time was set according to the Leksell GammaPlan.

Statistical Analysis

For statistical analysis three main tumor locations were defined. For macular melanoma the distance of the tumor to the fovea was less than 2 disc diameters, and for juxtapapillary melanoma the distance to the optic disc was less than 2 disc diameters; all other melanomas were classified as peripheral. The two groups (with and without radiation retinopathy) were compared by use of Fisher's exact test for categorical variables and the Wilcoxon test for quantitative variables. A *P* value of 0.05 or less was considered significant.

Results

The median follow-up period for all 32 patients was 38 months (range, 6–81 months). Twenty-five patients were followed on a regular basis over a median period of 46 months (range, 24–81 months). Seven patients (22%) required enucleation after a median interval of 13 months (range, 6–41 months) because of neovascular glaucoma in six cases and tumor recurrence in one (after 30 months). Two patients died of metastatic disease 38 and 61 months after treatment, respectively. Patients and tumor baseline characteristics are listed in Table 1. The mean age of the patients was 68 years (range, 37–93 years); 44% were male, 56% female. There was a history of diabetes mellitus in 3% and systemic hypertension in 19% of the patients; 12% had both. The baseline visual acuity was 20/20 to 20/40 in 28%, 20/50 to 20/200 in 50%, and 20/400-light perception in 22% of the cases. After the follow-up period, 44% of the patients experienced a vision loss to no light perception, 34% had a visual acuity of 20/400 to light perception, 19% of 20/50 to 20/200, and only one patient (3%) was better than 20/40.

Choroidal melanomas were located in the macular area in 53% of the patients, juxtapapillary in 22%, and in the periphery in 25%. The median largest basal tumor diameter was 11.11 mm (range, 4.9–21.4 mm), and the median tumor thickness was 5.16 mm (range, 2.12–11.51 mm). A median marginal dose of 50 Gy (range, 40–80 Gy) and mean maximal dose of 90 Gy (range, 70–125 Gy) were used for gamma knife treatment. There was one case of recurrence.

Overall, 27 (84%) of the 32 patients in our series developed radiation retinopathy during the follow-up period. The median time of onset after radiation was 8 months (range, 1–22 months). The earliest and most common finding was intraretinal hemorrhage, which was present in 70% of the patients with radiation retinopathy after a median interval of 8 months (range, 1–32 months). The second most common findings were macular edema and capillary nonperfusion in 63% of the cases, macular edema after a median period of 9 months (range, 1–21 months) and capillary nonperfusion after 10 months (range, 2–46 months). Lipid exudates developed in 53% of the patients after 9.5 months (range, 1–39 months). Cotton wool spots were found in 44% after

Table 1. Demographic and Tumor Baseline Characteristics in Patients with Choroidal Melanoma*

Variable	Radiation Retinopathy	No Radiation Retinopathy
Patients	27 (100)	5 (100)
Median age, yrs	67	68
Range	37–93	66–90
Gender		
Male	11 (41)	3 (60)
Female	16 (59)	2 (40)
Diabetes mellitus	5 (19)	0
Systemic hypertension	9 (33)	1 (20)
Median tumor thickness, mm	5.32	3.72
Range	2.1–11.5	2.5–11.2
Median largest tumor base, mm	12.58	9.9
Range	4.9–20.9	7.3–10.4
Location		
Macular	14 (52)	3 (60)
Peripapillary	7 (26)	0
Peripheral	6 (22)	2 (40)
Treatment dosage, Gy		
Median maximal dose	90	90
Median marginal dose	50	50
Follow-up, months	38	46
Range	6–81	24–76

*Unless otherwise indicated, data are given as number (percentage) of patients.

a median period of 9.5 months (range, 1–22 months). Microvascular degeneration caused by radiation retinopathy (e.g., microaneurysms) developed in only 30% of patients after a median period of 11.5 months (range, 5–27 months). Proliferative radiation retinopathy with development of optic disc or retinal neovascularization was found in 22% of all patients after a median period of 15 months (range, 5–30 months). Iris neovascularization occurred in 50% of patients after 11 months (range, 6–46 months) and progressed to neovascular glaucoma in 47% of all patients after a median postradiation period of 14 months (range, 6–46 months).

No significant correlation was found between the incidence of radiation retinopathy and the applied doses (Fig 1). Size of the tumor base, tumor thickness, location of the tumor had predictive value for radiation retinopathy. Only choroidal melanomas with a large basal diameter were more likely to develop radiation retinopathy ($P = 0.109$) (Fig 2).

Discussion

This retrospective study demonstrates that radiation retinopathy is a frequent complication after gamma knife treatment of choroidal melanomas with a marginal dosage of 40 to 80 Gy. The incidence of radiation retinopathy after a mean follow-up of 44 months was 84%. Radiation retinopathy was associated with neovascular glaucoma in 47% and blindness in 44% of all treated patients. Six eyes (19%) were enucleated because of severe radiation-induced complications such as neovascular glaucoma.

On the basis of the pioneering work done by Leksell, the gamma knife was used to treat brain tumors as an alternative to conventional surgical removal.¹³ The gamma knife is a self-contained unit with 201 cobalt-60 sources arranged in a hemispheric array such that the emitted beams of radiation reach a common point of intersection. The beams of 201 gamma radiation sources meet in a so-called cross firing, which causes a sharp decrease of the isodose. This hemispheric array of sources, the large number of small-diameter beams, and the steep dose gradients surrounding a targeted lesion make the physical characterization of the radiation field complex. The gamma knife delivers very precisely a specified dose of radiation to a small target, minimizing the effects on surrounding tissues. Because of these physical properties, the gamma knife would seem to be a suitable instrument for the treatment of intraocular tumors. Stereotactic radiosurgery is usually applied in a single fraction and can hardly be compared with such other techniques as proton, helium ion radiotherapy, or brachytherapy. Zehetmayer and coworkers¹⁴ did not find that there was any benefit when the dose was administered in one, two, or three fractions.

Stereotactic radiosurgery normally requires immobilization of the target, which, with intracranial tumors, is usually achieved with a stereotactic frame. Intraocular tumors additionally require suturing of the rectus muscles or a suction device.¹⁵ Because of the duration of a gamma knife procedure, including frame fixation, magnetic resonance imaging, treatment planning, and radiation, a retrobulbar block alone would not guarantee immobilization of the globe. Even after

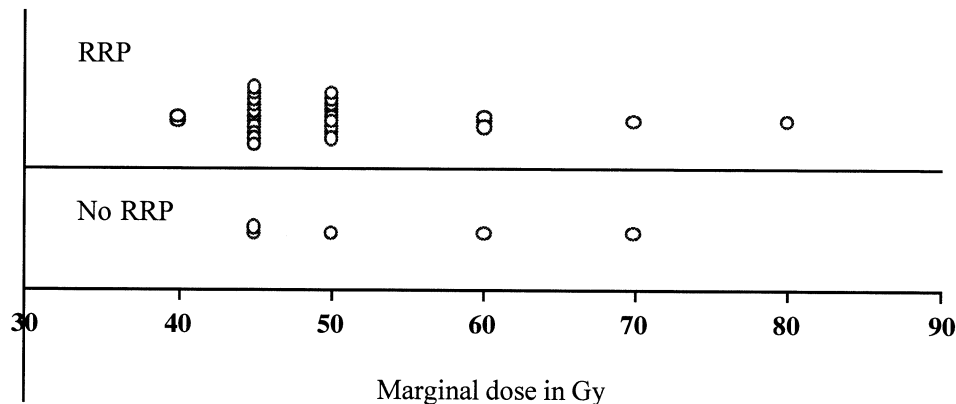


Figure 1. Patients with and without radiation retinopathy in correlation to the applied marginal dose. The marginal dose did not significantly correlate with radiation retinopathy risk.

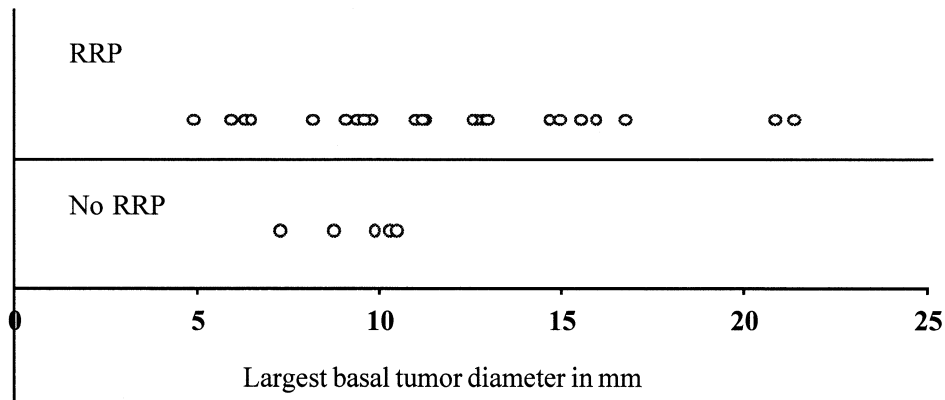


Figure 2. Patients with and without radiation retinopathy in correlation with largest basal tumor diameter. Although the correlation was not statistically significant, choroidal melanomas with a large basal diameter were more likely to develop radiation retinopathy ($P = 0.109$).

a complete retrobulbar block, eye movements can be seen during treatment after 1 hour or longer.

Magnetic resonance imaging can distinguish the serous and the solid parts of the tumor, which probably minimizes the treatment volume. Magnetic resonance imaging can visualize the lateral borders of small tumors better than computed tomography. Unlike proton beam radiotherapy, radioactive markers are not necessary.

In a rabbit eye model Rand et al¹⁶ demonstrated the ability of the gamma knife to destroy choroidal melanomas with a dose of 60 to 90 Gy at the 90% isodose. No viable tumor cells were found at autopsy in six of nine rabbits after 2 years. Chinela et al¹⁷ and Langmann et al⁴ were among the first to treat human choroidal melanomas with the gamma knife. They reported good tumor control in most cases with a marginal dose between 50 and 90 Gy, but gave no information on side effects. Rennie et al¹⁸ reported a regression of the choroidal melanoma in 13 of 14 patients after a single fraction gamma knife treatment with a dose of 70 Gy to the periphery of the tumor. After a mean follow-up of 23.5 months, 13 patients developed significant radiation-induced adverse reactions with radiation retinopathy in 11 cases (79%), iris neovascularization in 6 (43%), and neovascular glaucoma in 5 (36%). Although their 70-Gy dose was significantly higher than our mean dose of 50 Gy and their follow-up period was shorter, the incidence of radiation-induced complications is comparable to ours.

Radiation retinopathy is a serious and vision-threatening complication of radiotherapy for choroidal melanomas, whether charged particle or plaque treatment is applied. The reported incidence of radiation retinopathy varies from 10% to 89% and is related to a variety of factors including the type and dosage of radiation and the size and the location of the tumor.^{7-9,19,20} Although radiation retinopathy is found in 87% of patients after plaque radiotherapy for juxtapapillary melanomas and radiation maculopathy in 89% after proton beam irradiation, neither of those radiotherapies is as highly associated as gamma knife treatment (median marginal dosage 50 Gy) with such severe sight-limiting complications as rubeosis and neovascular glaucoma.^{10,19} In a large retrospective study Gunduz et al⁸ found radiation retinopathy after plaque radiotherapy in 43.1% of the pa-

tients and neovascular glaucoma in 7.1%. The factors predictive for radiation retinopathy were posterior tumor location with margin near the foveola (<4 mm) and high dose rate to the tumor base (>260 cGy/hour). Packer et al⁹ and Lommatzsch and Kirsch²⁰ reported lower incidences of radiation retinopathy. In Packer's series, 23.4% of patients treated with iodine-125 plaque therapy developed radiation retinopathy and 10.9% neovascular glaucoma. Lommatzsch reported postradiation retinopathy in only 10% and neovascular glaucoma in 2% after 106Ru/106Rh plaque radiotherapy, wherein the average height of the tumors treated was less than 5 mm. Radiation-induced complications are more commonly seen after charged particle radiation therapy than after plaque radiotherapy. In a prospective study Char et al²¹ compared the results of treating small and medium-sized tumors with either iodine-125 plaque or charged particle therapy using helium ions. The complication rate was significantly higher for the helium ion therapy than for the plaque therapy. The incidence for neovascular glaucoma was 11.2% in the plaque group versus 29.1% in the helium ions group. A similar neovascular glaucoma rate of 30.5% was obtained when melanomas with a mean tumor thickness of 6.5 mm were treated with helium ions in a retrospective study by the same group.²²

Foss and coworkers²³ reported on predictive factors for the development of rubeosis after proton beam radiotherapy for choroidal melanomas. This study involved tumors too large or too close to the optic nerve for plaque radiotherapy. They found a rubeosis rate of 34%, and enucleation was necessary because of neovascular glaucoma in 13%. Large tumor size and the presence of retinal detachment were significant risk factors for rubeosis.

In our study most of the melanomas were located either within 2 disc diameters of the fovea (53%) or in the juxtapapillary region (22%). We also treated more large tumors (median, 11.8 mm). These factors may be the reason for the high incidence of radiation retinopathy and of neovascular glaucoma in our study.

In another study, color Doppler ultrasonography showed that blood flow velocity in the central retinal artery was significantly reduced 2 years after stereotactic radiotherapy using the gamma knife.²⁴ That study used a marginal dose

of 41 to 66.5 Gy, similar to ours. This reduction in blood flow could also be responsible for radiation retinopathy.

All diabetic patients in our series developed radiation retinopathy, and this could have contributed to our high incidence of retinal complications. The clinical findings and the time of onset of radiation retinopathy or of neovascular glaucoma did not differ from those of nondiabetic patients.

In conclusion, single-fraction Leksell gamma knife treatment of choroidal melanomas with a median marginal dosage of 50 Gy is highly associated with early radiation retinopathy and with neovascular glaucoma. Nearly half of the patients experienced a complete loss of vision as a result of this high complication rate. Although no significant main predictor of radiation retinopathy was found, eyes with a large basal tumor diameter were more likely to develop this retinal complication after gamma knife treatment.

References

1. Seddon JM, Gragoudas ES, Egan KM, et al. Relative survival rates after alternative therapies for uveal melanoma. *Ophthalmology* 1990;97:769–77.
2. Shields CL, Shields JA, Cater J, et al. Plaque radiotherapy for uveal melanoma: long-term visual outcome in 1106 consecutive patients. *Arch Ophthalmol* 2000;118:1219–28.
3. Damato B, Groenewald C, McGalliard J, Wong D. Endoresection of choroidal melanoma. *Br J Ophthalmol* 1998;82:213–8.
4. Langmann G, Pendl G, Schröttner O, et al. Die radiochirurgische Therapie mit der Leksell Gamma Einheit in der Behandlung von Aderhautmelanomen. *Spektr Augenheilkd* 1995;9:16–21.
5. Oosterhuis JA, Journee-de Korver HG, Kakebeeke-Kemme HM, Bleeker JC. Transpupillary thermotherapy in choroidal melanomas. *Arch Ophthalmol* 1995;113:315–21.
6. Zambrano AD, Chinela A, Bunge H, et al. Radiocirugía estereotóxica de los melanomas uveales. *Protocolo de tratamiento*. *Arch Oftalmol* 1989;64:49–56.
7. Gragoudas ES. Jules Gonin Lecture of the Retina Research Foundation. Long-term results after proton irradiation of uveal melanomas. *Graefes Arch Clin Exp Ophthalmol* 1997;196:235:265–7.
8. Gunduz K, Shields CL, Shields JA, et al. Radiation retinopathy following plaque radiotherapy for posterior uveal melanoma. *Arch Ophthalmol* 1999;117:609–14.
9. Packer S, Stoller S, Lesser ML, et al. Long-term results of iodine 125 irradiation of uveal melanoma. *Ophthalmology* 1992;99:767–73; discussion 774.
10. De Potter P, Shields C, Shields JA, et al. Plaque radiotherapy for juxtapapillary choroidal melanoma. Visual acuity and survival outcome. *Arch Ophthalmol* 1996;114:1357–65.
11. Amoaku WMK, Archer DB. Cephalic radiation and retinal vasculopathy. *Eye* 1990;4:195–203.
12. Viebahn M, Barricks ME, Osterloh MD. Synergism between diabetic and radiation retinopathy: case report and review. *Br J Ophthalmol* 1991;75:629–32.
13. Leksell L. Stereotactic radiosurgery. *J Neurol Neurosurg Psychiatry* 1983;46:797–803.
14. Zehetmayer M, Kitz K, Menapace R, et al. Local tumor control and morbidity after one to three fractions of stereotactic external beam irradiation for uveal melanoma. *Radiother Oncol* 2000;55:135–44.
15. Zehetmayer M, Menapace R, Kitz K, Ertl A. Suction attachment for stereotactic radiosurgery of intraocular malignancies. *Ophthalmologica* 1994;208:119–21.
16. Rand RW, Khonsary A, Brown WJ, et al. Leksell stereotactic surgery in the treatment of eye melanoma. *Neurol Res* 1987;9:142–6.
17. Chinela AB, Zambrano AD, Bunge HJ. Gamma knife radiosurgery in uveal melanomas. In: Steiner L, ed. *Radiosurgery: Baseline and Trends*. New York: Raven Press, 1992;161–9.
18. Rennie I, Forster D, Kemeny A, et al. The use of single fraction Leksell stereotactic radiosurgery in the treatment of uveal melanoma. *Acta Ophthalmol Scand* 1996;74:558–62.
19. Guyer DR, Mukai S, Egan KM, et al. Radiation maculopathy after proton beam irradiation for choroidal melanoma. *Ophthalmology* 1992;99:1278–85.
20. Lommatzsch PK, Kirsch IH. 106Ru/106Rh plaque radiotherapy for malignant melanomas of the choroid. With follow-up results more than 5 years. *Doc Ophthalmol* 1988;68:225–38.
21. Char DH, Quivey JM, Castro JR, et al. Helium ions versus iodine 125 brachytherapy in the management of uveal melanoma. A prospective, randomized, dynamically balanced trial. *Ophthalmology* 1993;100:1547–54.
22. Char DH, Castro JR, Kroll S, et al. Five-year follow-up of helium ion therapy for uveal melanoma. *Arch Ophthalmol* 1990;108:209–14.
23. Foss AJE, Whelehan I, Hungerford JL, et al. Predictive factors for the development of rubeosis following proton beam radiotherapy for uveal melanoma. *Br J Ophthalmol* 1997;81:748–54.
24. Vecsei PV, Kirchner K, Nagel G, et al. Ocular arterial blood flow of choroidal melanoma eyes before and after stereotactic radiotherapy using Leksell gamma knife: 2 year follow up. *Br J Ophthalmol* 1999;83:1324–8.